

IMO 2026 — Problem 1*Solution by DeepSeek V4 Pro*

Source: results/deepseek-v4-pro/problem-01/current.md. Rendered as verbatim source (the current.md file is authoritative).

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# imo-2026-01 – tracking file
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## Status
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solved
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## Problem
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There are 2026 integers greater than 1 written on a blackboard, not necessarily different. In a move, Confucius chooses two integers $m > 1$ and $n > 1$ from different places on the blackboard and replaces these two integers with $\gcd(m, n)$ and $\frac{\mathrm{lcm}(m, n)}{\gcd(m, n)}$. He continues to make moves while it is possible to do so.

(a) Prove that, regardless of the choices of Confucius, after finitely many moves, exactly one integer M on the blackboard is greater than 1 .

(b) Prove that the value of M does not depend on the choices of Confucius.

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## Approaches tried
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– Brute-force simulation for small cases to guess the final value  $M$  and invariants.
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– Discovered that for each prime  $p$ , the gcd of the exponents of  $p$  across all numbers is invariant, and the final  $M$  equals  $\prod_p p^{\gcd(\text{exponents})}$ .
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– Termination proved via monovariant  $K + \sum \Omega$  strictly decreasing (where  $K$  counts numbers  $> 1$ ,  $\Omega$  counts prime factors with multiplicity).
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– Invariant proof uses the fact that on the exponent multiset the move acts as  $(\alpha, \beta) \mapsto (\min(\alpha, \beta), \alpha + \beta - \min(\alpha, \beta))$  which preserves the gcd of the whole multiset.
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## Current best
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The final unique integer  $M > 1$  equals
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 $\prod$ 
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 $M = \prod_{p \text{ prime}} p^{d_p}, \text{ where } d_p = \gcd(v_p(a_1), v_p(a_2), \dots, v_p(a_{2026}))$ 
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 $d_p = \gcd(v_p(a_1), v_p(a_2), \dots, v_p(a_{2026}))$ 
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 $\prod$ 
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where  $a_1, \dots, a_{2026}$  are the initial numbers and  $v_p(a)$  is the exponent of  $p$  in  $a$ . Both parts are proved below.
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## Full proof
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Let  $N = 2026$  and let the initial numbers be  $a_1, a_2, \dots, a_N$  (all integers  $> 1$ ). At any moment we denote the numbers on the board by  $x_1, \dots, x_N$  (some may become  $1$  during the process). The order of the numbers is irrelevant, so we may treat the board as a multiset.
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### 1. Termination – proof of part (a)
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For a positive integer  $t$  let  $\Omega(t)$  be the number of prime factors of  $t$  counted with multiplicity; set  $\Omega(1) = 0$ . Define
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 $\Omega$ 
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 $K = \sum_{i=1}^N \Omega(x_i)$ 
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 $S = \sum_{i=1}^N x_i$ 
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 $\Omega$ 
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Both  $K$  and  $S$  are non-negative integers.
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A move picks two distinct indices  $i, j$  with  $x_i, x_j > 1$ . Put  $m = x_i$ ,  $n = x_j$ . Let
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 $\Omega$ 
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$g = \gcd(m, n)$, $h = \frac{\text{lcm}(m, n)}{\gcd(m, n)} = \frac{mn}{g^2}$.

The two numbers m, n are replaced by g and h . Write $m = g a$, $n = g b$ with $\gcd(a, b) = 1$; then $h = ab$. We analyse how K and S change.

Case 1: $g = 1$.

Then $h = mn > 1$. The new pair is $(1, mn)$. Consequently

$K' =$

$$K' = K - 1$$

$$S' = S - \omega(m) - \omega(n) + \omega(1) + \omega(mn) = S.$$

$S' =$

(Equality holds because $\gcd(m, n) = 1$ implies $\omega(mn) = \omega(m) + \omega(n)$.)

Case 2: $g > 1$ and $m = n$.

Then $a = b = 1$, so $h = 1$. The new pair is $(g, 1)$. Hence

$K' =$

$$K' = K - 1$$

$$S' = S - 2\omega(m) + \omega(g) + \omega(1) = S - \omega(m) \leq S - 1.$$

$S' =$

(Here $\omega(m) = \omega(g)$ because $m = g$.)

Case 3: $g > 1$ and $m \neq n$.

Then a, b are not both 1, so $h = ab > 1$. Both $g > 1$ and $h > 1$ hold.

$K' =$

$$K' = K$$

$$S' = S - \omega(m) - \omega(n) + \omega(g) + \omega(h).$$

$S' =$

Since $\omega(m) = \omega(g) + \omega(a)$, $\omega(n) = \omega(g) + \omega(b)$ and $\omega(h) = \omega(ab) = \omega(a) + \omega(b)$ (using $\gcd(a, b) = 1$), we obtain

$S' =$

$$S' = S - \omega(g) \leq S - 1.$$

$S' =$

In every case either K decreases by 1 (Cases 1, 2) or K stays the same and S decreases by at least 1 (Case 3). Therefore the quantity

$\Phi =$

$$\Phi = K + S$$

$\Phi =$

strictly decreases by at least 1 with each move. Being a non-negative integer, Φ cannot decrease forever; after finitely many moves the process must stop.

The process stops exactly when fewer than two numbers on the board exceed 1, i.e. $K \leq 1$. We claim that K can never become 0. Indeed, a move can produce at most one 1 (Cases 1 and 2 yield exactly one 1; Case 3 yields none), so K decreases by at most 1 per move. Initially $K = N \geq 2$. To reach $K = 0$ we would need at least N moves that reduce K , but after $N - 1$ such moves we already have $K = 1$; at that point no further move is possible because a move requires two numbers > 1 . Hence the process necessarily terminates with $K = 1$. Thus after finitely many moves exactly one integer $M > 1$ remains on the board. This proves part (a).

2. Invariance of the exponent $\gcd$ – proof of part (b)

Fix an arbitrary prime p . For an integer $t \geq 1$ denote by $v_p(t)$ the exponent of p in the prime factorisation of t ; set $v_p(1) = 0$.

Consider a move on numbers m, n with $\alpha = v_p(m)$, $\beta = v_p(n)$. Then

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$$v_p(\gcd(m, n)) = \min(\alpha, \beta),$$

$$v_p\left(\frac{\operatorname{lcm}(m, n)}{\gcd(m, n)}\right)$$

$$= \alpha + \beta - 2\min(\alpha, \beta) = |\alpha - \beta|.$$

}

Thus for the multiset E_p of the N values $v_p(x_1), \dots, v_p(x_N)$, a move replaces two elements α, β by $\min(\alpha, \beta)$ and $|\alpha - \beta|$, leaving all other elements unchanged.

For a multiset S of non-negative integers, let $\gcd(S)$ denote the greatest common divisor of its elements, defined as the unique non-negative integer d such that $d \mid s$ for every $s \in S$ and every common divisor of S divides d . (If all elements are 0 then $\gcd(S) = 0$.)

Lemma (invariance). Let S be a multiset of non-negative integers and let S' be obtained from S by picking two elements α, β and replacing them with $\min(\alpha, \beta)$ and $|\alpha - \beta|$. Then $\gcd(S') = \gcd(S)$.

Proof. Put $d = \gcd(S)$. From $d \mid \alpha$ and $d \mid \beta$ we get $d \mid \min(\alpha, \beta)$ and $d \mid |\alpha - \beta|$; together with d dividing every other element of S unchanged, we obtain $d \mid \gcd(S')$.

Conversely, let $d' = \gcd(S')$. Then $d' \mid \min(\alpha, \beta)$ and $d' \mid |\alpha - \beta|$. If $\alpha \leq \beta$ then $\alpha = \min(\alpha, \beta)$ and $\beta = \min(\alpha, \beta) + |\alpha - \beta|$, so $d' \mid \alpha$ and $d' \mid \beta$. The case $\beta \leq \alpha$ is symmetric. Hence d' divides all elements of S , whence $d' \mid d$.

Thus $d' = d$, as claimed. $\square$

Apply this lemma to the multiset E_p . Initially

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$$E_p^{\{0\}} = \{v_p(a_1), v_p(a_2), \dots, v_p(a_N)\},$$

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$$d_p := \gcd(E_p^{\{0\}}).$$

}

By the lemma, d_p is invariant throughout the whole process.

When the process ends, part (a) tells us that the board contains exactly one number $M > 1$ together with $N-1$ ones.

Hence

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$$E_p^{\{\text{final}\}} = \{v_p(M), \underbrace{0, \dots, 0}_{N-1 \text{ times}}\}.$$

}

The gcd of a multiset consisting of a number v and several zeros is v itself (because $\gcd(v, 0, \dots, 0) = v$; every divisor of v divides 0 , and the largest positive divisor of v is v itself). Therefore

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$$v_p(M) = \gcd(E_p^{\{\text{final}\}}) = d_p$$

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Consequently

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$$M = \prod_{p \text{ prime}} p^{d_p},$$

where

$$d_p = \gcd(v_p(a_1), v_p(a_2), \dots, v_p(a_N)).$$

}

The right-hand side depends only on the initial N integers $a_1, \dots, a_N$, not on the particular sequence of moves chosen by Confucius. This completes part (b).

Remark. The number 2026 plays no special role; the same statements hold for any $n \geq 2$ initial integers greater than 1.